

Executive Summary

Workforce Attrition Patterns and Risk Hotspot Analysis at Palo Alto Networks

1. Overview

Employee attrition can create challenges related to workforce continuity, knowledge retention, recruitment costs, and workforce planning. This project analyzes employee attrition patterns to identify workforce segments with comparatively higher observed attrition rates and highlight areas that may require further investigation.

The analysis uses a dataset containing **1,470 employee records and 31 variables**, covering demographic, organizational, employment, workload, travel, and commuting characteristics. The project adopts a **diagnostic analytics approach**, focusing on identifying patterns and attrition hotspots rather than predicting individual employee departures.

2. Key Findings

The overall observed attrition rate was **16.12%**, with **237 employees exiting** and **1,233 employees retained**.

Several workforce segments showed notably higher observed attrition:

Area	Key Finding
Department	Sales had the highest departmental attrition at 20.63%
Job Role	Sales Representatives had the highest observed attrition at 39.76%
Age	Employees aged 18–25 recorded 35.77% attrition
Tenure	Employees with 0–2 years at the company recorded 29.82% attrition
Overtime	Employees working overtime recorded 30.53% attrition vs. 10.44% without overtime

Area	Key Finding
Business Travel	Frequent travellers recorded 24.91% attrition vs. 8.00% among non-travellers
Distance	Employees living 21+ units from work recorded 22.06% attrition vs. 13.77% for 0–5
Gender	Male employees: 17.01% ; Female employees: 14.80%

These findings indicate that attrition is **not evenly distributed across the workforce**. Certain combinations of role, tenure, workload, age, travel, and commuting characteristics represent comparatively higher observed attrition segments.

3. Major Attrition Hotspots

Three areas stand out for further workforce investigation:

Early-Tenure Employees

Employees with less than three years at the organization showed substantially higher observed attrition than longer-tenured employees. This suggests that **onboarding, role clarity, career development, mentoring, and early employee experience** should be examined.

Workload and Overtime

The difference between employees working overtime and those who did not was substantial. Teams with consistently high overtime may therefore warrant investigation into **workload distribution, staffing levels, scheduling, and work-life balance**.

Specific Roles and Workforce Segments

Sales Representatives recorded the highest observed job-role attrition. Rather than treating the role itself as a cause, further investigation should examine factors such as **job demands, performance expectations, career progression, compensation, and workload** within the role.

4. Recommended Areas of Action

Based on the analysis, stakeholders should consider:

1. **Strengthening early-tenure support** through structured onboarding, mentoring, and regular employee check-ins.
2. **Investigating high-overtime teams** to understand workload and staffing pressures.
3. **Reviewing high-attrition job roles** for role-specific workload, career progression, and employee experience factors.
4. **Examining travel and commuting patterns** alongside workload and work-life balance.
5. **Monitoring attrition continuously** across departments, roles, tenure groups, and other workforce segments.
6. **Combining multiple workforce dimensions** rather than relying only on the overall attrition percentage.

5. Business Intelligence Solution

An interactive **Streamlit dashboard** was developed using **Python, Pandas, and Plotly** to make the analysis accessible to stakeholders.

The dashboard provides:

- Overall attrition KPIs
- Retained vs. exited employee analysis
- Department and job-role hotspots
- Age, gender, and education analysis
- Tenure analysis
- Overtime analysis
- Business-travel analysis
- Distance-from-home analysis
- Interactive filters for department, job role, tenure, overtime, and business travel

This allows stakeholders to move from an organization-wide view to specific workforce segments and investigate where attrition is concentrated.

6. Conclusion

The analysis shows that employee attrition varies considerably across workforce segments. **Early tenure, overtime, younger age groups, selected job roles, frequent business travel, and greater distance from the workplace** were associated with comparatively higher observed attrition in the dataset.

The findings should be treated as **diagnostic signals rather than evidence of causation**. The analysis provides a starting point for HR and workforce-management teams to investigate underlying factors and design targeted retention initiatives.

The combination of **exploratory analytics, an interactive dashboard, and stakeholder-oriented recommendations** provides a practical framework for monitoring workforce attrition and identifying areas requiring deeper investigation.